CT 210 CASE STUDY

Chikamso Ezeaku, Richard Amoako

Instructor Name: Joel Wiredu

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Project Overview/ Client Overview ABC Corporation, an established player in the consumer services market, required a comprehensive upgrade and expansion of its existing network infrastructure to accommodate a new department and improve overall operational efficiency. The key requirements outlined by the client included enhanced security measures, robust data handling capabilities, and improved connectivity across departments.

Project Objective: Our firm was tasked with designing and implementing a state-of-the-art network that not only meets the current demands of ABC Corporation but also provides scalability for future expansions. The project aimed to replace the outdated network components, optimize data flow, and ensure secure access to network resources.

Network Topology

The proposed network topology is a simulation model. Different switches connect to various departmental nodes, with inter-department connections to ensure uninterrupted data exchange even in case of a node failure. For detailed visuals, refer to Appendix A: Network Topology.

IP Address Design

Our network design incorporates Variable Length Subnet Masking (VLSM) to maximize the efficient use of IP addresses. The design ensures that each department has sufficient IP addresses for current and future needs while minimizing wastage. Specific IP address configurations and subnetting details are available in Appendix B: IP Address Design.

Network Services

This section includes services configured on the network. The following has been configured on the network:

- Hostnames for the three switches and router
- Security:
  - Password
  - Encryptions
  - SSH
- Banner MOTD
- IP Addressing

To view the configs of each device on the network refer to Appendix C: Running and Starting Configs of each Intermediary Device.

Hostname

The purpose of assigning hostnames to the switch was to make for easier identification and management.

Switch Subnet A – Subnet A

Switch Subnet B – Subnet B

Subnet Switch C – Subnet C

Router – NetProEdge

Security

The following section explains the different security features established on the network.

Passwords

For the console ‘class123’ was assigned

For the Exec ‘cisco123’ was assigned

SSH

SSH is a secure protocol for remote access ensuring encrypted communication. Configuring a domain, username, and password is essential for authentication and security. VTY lines allow for remote access. Kindly refer to **Appendix D: SSH Connectivity** to view successful ssh connection to each intermediary device.

Additional Security Measures

- Implemented multiple layers of security.
- Password requirements for all devices
- Automatic timeouts on the console and VTY lines to prevent unauthorized
- Data encryption across the network to protect sensitive information.

Banner Message of the Day

The purpose of the banner MOTD is to warn users of unauthorized access being prohibited.

IP Addressing

IP addressing assigns unique IP's to devices within a network, enabling communication. It helps routers and switches route data effectively. For detailed configurations on each device, refer to **Appendix E: IP Summary Brief**. To verify connectivity, refer to **Appendix F: IP Connectivity** for testing results.

MAC Address Table

A MAC address is a unique identifier for each network device, used to facilitate local communication. Switches store these addresses in the MAC address table, ensuring correct data forwarding. To view the MAC address tables for all switches, please refer to **Appendix G: Switch MAC Address Tables.**

ARP Table

ARP maps IP addresses to MAC addresses, enabling network devices to identify each other on a local network. This protocol ensures data packets are directed accurately. To view the ARP table of the NetProEdge router, refer to **Appendix H: NetProEdge ARP Table.**

Routing Table

The routing table is essential for directing packets between different networks. It contains routes, destinations, and next hops. The routing table is critical for network efficiency and proper data flow. For the NetProEdge routing table, refer to **Appendix I: NetProEdge Routing Table** for details.

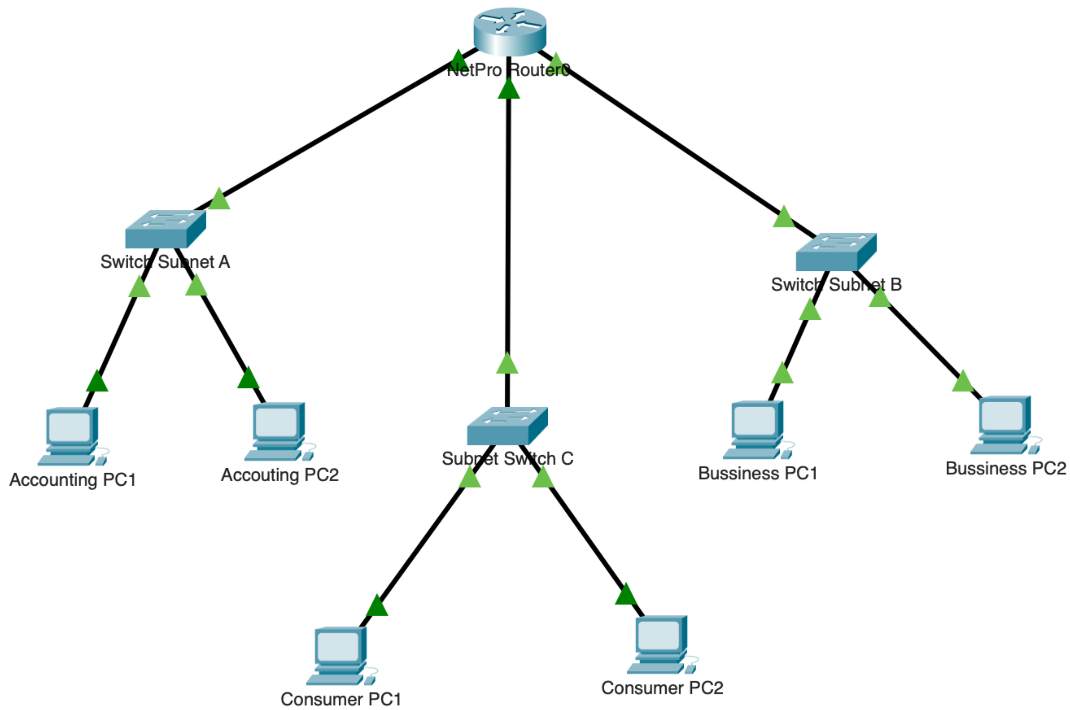
Packet Tracer Implementation

Used Cisco Packet Tracer as it's a network simulation tool used to design and test network topologies. It allows configuration validation and troubleshooting in a controlled environment. The Packet Tracer file simulating the network has been submitted with the RFP for further review and testing.

Conclusion

The newly implemented network has met all the specified requirements of ABC Corporation, offering a secure, and efficient networking environment. The project has been documented comprehensively to assist in future expansions or troubleshooting.

Appendix A: Network Topology



Appendix B: IP Address Design

Subnet	**Department**	**Number of Hosts**
Subnet A	Accounting	500
Subnet B	Business Services	100
Subnet C	Customer Services	30

Subnet A

Network Address	Subnet Mask	First Host address	Last Host address	Broadcast
172.17.168.0/23	255.255.254.0	172.17.168.1	172.17.169.254	172.17.169.255

Subnet B

Network Address	Mask	First Host address	Last Host address	Broadcast
172.17.170.0/25	255.255.255.128	172.17.170.1	172.17.170.126	172.17.170.127

Subnet C

Network Address	Mask	First Host address	Last Host address	Broadcast
172.17.170.128/27	255.255.255.224	172.17.170.129	172.17.170.158	172.17.170.159

Subnet C

Office on the web Frame

Network Address	Mask	First Host address	Last Host address	Broadcast
172.17.170.128/27	255.255.255.224	172.17.170.129	172.17.170.158	172.17.170.159

Device	Network Address	IP address	Subnet Mask	Default Gateway
NetPro Router ISP Lo0	116.205.0.16/30	116.205.0.17	255.255.255.252	116.205.0.17
NetPro Router Gi 0/0	172.17.168.0/23	172.17.168.1	255.255.254.0	172.17.168.1
Switch Subnet A VLAN 1	172.17.168.0/23	172.17.168.2	255.255.254.0	172.17.168.1
Host 1 Subnet A	172.17.168.0/23	172.17.168.11	255.255.254.0	172.17.168.1
Host 2 Subnet A	172.17.168.0/23	172.17.168.12	255.255.254.0	172.17.168.1
NetPro Router Gi 0/1	172.17.170.0/25	172.17.170.1	255.255.255.128	172.17.170.1
Switch Subnet B VLAN 1	172.17.170.0/25	172.17.170.2	255.255.255.128	172.17.170.1
Subnet B Host 1	172.17.170.0/25	172.17.170.11	255.255.255.128	172.17.170.1
Subnet B Host 2	172.17.170.0/25	172.17.170.12	255.255.255.128	172.17.170.1

Subnet C

Office on the web Frame

NetPro Router Gi 0/2	172.17.170.0/25	172.17.171.1	255.255.255.128	172.17.171.1
Switch Subnet C VLAN 1	172.17.171.0/25	172.17.171.2	255.255.255.128	172.17.171.1
Subnet C Host 1	172.17.171.0/25	172.17.171.11	255.255.255.128	172.17.171.1
Subnet C Host 2	172.17.171.0/25	172.17.171.12	255.255.255.128	172.17.171.1

Appendix C: Running and Starting Configurations of Each Intermediary Device

Subnet A - Switch 1 Startup-Configs

```
SubnetA#show start
Using 1476 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname S1
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
no ip domain-lookup
ip domain-name netpro.com
!
username cisco secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
```

```
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
description Management VLAN
ip address 172.17.168.2 255.255.254.0
!
ip default-gateway 172.17.168.1
!
banner motd ^CUnauthorized Access is Prohibited!^C
!
!
!
line con 0
password 7 0822404F1A0A544541
logging synchronous
login
!
line vty 0 4
exec-timeout 0 4
```

```
logging synchronous
login local
transport input ssh
line vty 5 15
login
!
!
!
!
End
```

Subnet B Switch 1 Startup-Configs

```
SubnetB#show start

Using 1459 bytes

!

version 15.0

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname S2

!

enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/

!

!

!

no ip domain-lookup

ip domain-name netpro.com

!

username cisco secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1

!

!

!
```

```
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
```

```
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
description Management Vlan
ip address 172.17.170.2 255.255.255.128
!
ip default-gateway 172.17.170.1
```

```
!
banner motd ^CUnauthorized Access is Prohibited^C
!
!
!
line con 0
password 7 0822404F1A0A544541
logging synchronous
login
!
line vty 0 4
logging synchronous
login local
transport input ssh
line vty 5 15
login
!
!
!
!
end
```

Subnet C Switch Startup-ConfigS

```
SubnetC#show start
Using 1502 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
```

```
!
hostname SubnetC
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name netpro.com
!
username cisco secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
```

```
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
```

```
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 172.17.170.129 255.255.255.224
!
ip default-gateway 172.17.170.129
!
banner motd ^CUnauthorized Access Prohibited!^C
!
!
!
line con 0
 password 7 0822404F1A0A544541
 logging synchronous
 login
!
line vty 0 4
 logging synchronous
 login local
 transport input ssh
line vty 5 15
 logging synchronous
```

```
 login local
 transport input ssh
!
!
!
!
end
```

NetProEdge Router Startup-Configs

```
NetProEdge#show start
Using 1245 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
security passwords min-length 8
!
hostname NetProEdge
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
username cisco privilege 15 secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
license udi pid CISCO2911/K9 sn FTX1524MUZN-
!
!
!
```

```
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name netpro.com
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface Loopback0
ip address 116.205.0.17 255.255.255.252
!
interface GigabitEthernet0/0
ip address 172.17.168.1 255.255.254.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 172.17.170.1 255.255.255.128
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 172.17.171.1 255.255.255.128
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
ip default-gateway 172.17.171.1
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^CUnathorized Access Prohibited!^C
```

```
!
!
!
!
line con 0
password 7 0822404F1A0A544541
login
!
line aux 0
!
line vty 0 4
logging synchronous
login local
transport input ssh
line vty 5 15
logging synchronous
login local
transport input ssh
!
!
!
end
```

Subnet A Switch 1 Running Configs

```
SubnetA#show run
Building configuration...

Current configuration : 1481 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname SubnetA
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
no ip domain-lookup
ip domain-name netpro.com
!
username cisco secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
!
```

```
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
```

```
description Management VLAN
ip address 172.17.168.2 255.255.254.0
!
ip default-gateway 172.17.168.1
!
banner motd ^CUnauthorized Access is Prohibited!^C
!
!
!
line con 0
password 7 0822404F1A0A544541
logging synchronous
login
!
line vty 0 4
exec-timeout 0 4
logging synchronous
login local
transport input ssh
line vty 5 15
login
!
!
!
!
end
```

Subnet B Switch 1 Running Configs

```
SubnetB#show run

Building configuration...


Current configuration : 1464 bytes

!

version 15.0

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname SubnetB

!

enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/

!

!
```

```
!
no ip domain-lookup
ip domain-name netpro.com
!
username cisco secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
```

```
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
description Management Vlan
ip address 172.17.170.2 255.255.255.128
```

```
!
ip default-gateway 172.17.170.1
!
banner motd ^CUnauthorized Access is Prohibited^C
!
!
!
line con 0
password 7 0822404F1A0A544541
logging synchronous
login
!
line vty 0 4
logging synchronous
login local
transport input ssh
line vty 5 15
login
!
!
!
!
end
```

Subnet C Switch Running Configs

```
SubnetC#show run
Building configuration...

Current configuration : 1498 bytes
!
version 15.0
```

```
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname SubnetC
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name netpro.com
!
username cisco secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
```

```
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
```

```
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
ip address 172.17.171.2 255.255.255.128
!
ip default-gateway 172.17.171.1
!
banner motd ^CUnauthorized Access Prohibited!^C
!
!
!
line con 0
password 7 0822404F1A0A544541
logging synchronous
login
!
line vty 0 4
logging synchronous
login local
transport input ssh
line vty 5 15
logging synchronous
login local
transport input ssh
!
!
!
!
```

```
end
```

NetProEdge Router Running Configs

```
NetProEdge#show run
Building configuration...

Current configuration : 1428 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
security passwords min-length 8
!
hostname NetProEdge
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
username cisco privilege 15 secret 5 $1$mERr$/VT.EosIwbboaDHzZXq7r1
!
!
license udi pid CISCO2911/K9 sn FTX1524MUZN-
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name netpro.com
!
!
spanning-tree mode pvst
!
!
!
!
!
```

```
!
interface Loopback0
description ISP Network Simulation
ip address 116.205.0.17 255.255.255.252
!
interface GigabitEthernet0/0
description Connection to Subnet A
ip address 172.17.168.1 255.255.254.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
description Connection to Subnet B
ip address 172.17.170.1 255.255.255.128
duplex auto
speed auto
!
interface GigabitEthernet0/2
description Connection to Subnet C
ip address 172.17.171.1 255.255.255.128
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
ip default-gateway 172.17.171.1
ip classless
ip route 0.0.0.0 0.0.0.0 116.205.0.18
!
ip flow-export version 9
!
!
!
banner motd ^CUnathorized Access Prohibited!^C
!
!
!
!
line con 0
password 7 0822404F1A0A544541
login
!
line aux 0
!
line vty 0 4
logging synchronous
login local
transport input ssh
line vty 5 15
logging synchronous
login local
transport input ssh
!
!
```

!
end

Appendix D: SSH Connectivity

Subnet A Switch 1 SSH Connectivity

```
C:\>ssh -l cisco 172.17.168.2

Password:

Unauthorized Access is Prohibited!

SubnetA>
```

Subnet B Switch 1 SSH Connectivity

```
C:\>ssh -l cisco 172.17.170.2

Password:

Unauthorized Access is Prohibited

SubnetB>!
```

Subnet C Switch 1 SSH Connectivity

```
C:\>ssh -l cisco 172.17.171.2

Password:

Unauthorized Access Prohibited!

SubnetC>!
```

NetProEdge Router SSH Connectivity

```
C:\>ssh -l cisco 116.205.0.17

Password:

Unathorized Access Prohibited!

NetProEdge#!!
```

Appendix E: IP Summary Brief

Subnet A IP Brief

```
SubnetA>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  172.17.168.2    YES manual up                    up
```

Subnet B IP Brief

```
SubnetB>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  172.17.170.2    YES manual up                    up
```

Subnet C IP Brief

```
SubnetC>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  172.17.171.2    YES manual up                    up
```

NetProEdge Router IP Brief

```
NetProEdge>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.17.168.1    YES manual up                    up
GigabitEthernet0/1     172.17.170.1    YES manual up                    up
GigabitEthernet0/2     172.17.171.1    YES manual up                    up
Loopback0              116.205.0.17    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

Appendix F: IP Connectivity

Subnet A PC 1 to Subnet A PC 2

```
C:\>ping 172.17.168.12

Pinging 172.17.168.12 with 32 bytes of data:

Reply from 172.17.168.12: bytes=32 time=2ms TTL=128
Reply from 172.17.168.12: bytes=32 time<1ms TTL=128
Reply from 172.17.168.12: bytes=32 time=1ms TTL=128
Reply from 172.17.168.12: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.168.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Subnet A PC 1 to Subnet B PC 1

```
C:\>ping 172.17.170.11

Pinging 172.17.170.11 with 32 bytes of data:

Reply from 172.17.170.11: bytes=32 time<1ms TTL=127
Reply from 172.17.170.11: bytes=32 time<1ms TTL=127
Reply from 172.17.170.11: bytes=32 time<1ms TTL=127
Reply from 172.17.170.11: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.170.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Subnet A PC 1 to Subnet B PC 2

```
C:\>ping 172.17.170.12

Pinging 172.17.170.12 with 32 bytes of data:

Reply from 172.17.170.12: bytes=32 time<1ms TTL=127
Reply from 172.17.170.12: bytes=32 time=1ms TTL=127
Reply from 172.17.170.12: bytes=32 time=7ms TTL=127
Reply from 172.17.170.12: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.170.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

Subnet A PC 2 to Subnet B PC

```
C:\>ping 172.17.170.11

Pinging 172.17.170.11 with 32 bytes of data:

Reply from 172.17.170.11: bytes=32 time<1ms TTL=127
Reply from 172.17.170.11: bytes=32 time<1ms TTL=127
Reply from 172.17.170.11: bytes=32 time<1ms TTL=127
Reply from 172.17.170.11: bytes=32 time=11ms TTL=127

Ping statistics for 172.17.170.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

Subnet A PC 2 to Subnet B PC 2

```
C:\>ping 172.17.170.12

Pinging 172.17.170.12 with 32 bytes of data:

Reply from 172.17.170.12: bytes=32 time<1ms TTL=127
Reply from 172.17.170.12: bytes=32 time<1ms TTL=127
Reply from 172.17.170.12: bytes=32 time<1ms TTL=127
Reply from 172.17.170.12: bytes=32 time<1ms TTL=127

Ping statistics for 172.17.170.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Subnet B PC 1 to Subnet B PC 2

```
Pinging 172.17.170.12 with 32 bytes of data:

Reply from 172.17.170.12: bytes=32 time<1ms TTL=128
Reply from 172.17.170.12: bytes=32 time<1ms TTL=128
Reply from 172.17.170.12: bytes=32 time<1ms TTL=128
Reply from 172.17.170.12: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.170.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Subnet A PC 1 to Subnet A Switch

```
C:\>ping 172.17.168.2

Pinging 172.17.168.2 with 32 bytes of data:

Reply from 172.17.168.2: bytes=32 time=1ms TTL=255
Reply from 172.17.168.2: bytes=32 time<1ms TTL=255
Reply from 172.17.168.2: bytes=32 time<1ms TTL=255
Reply from 172.17.168.2: bytes=32 time<1ms TTL=255

Ping statistics for 172.17.168.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Subnet A PC 1 to Subnet A Default Gateway

```
C:\>ping 172.17.168.1

Pinging 172.17.168.1 with 32 bytes of data:

Reply from 172.17.168.1: bytes=32 time=2ms TTL=255
Reply from 172.17.168.1: bytes=32 time<1ms TTL=255
Reply from 172.17.168.1: bytes=32 time<1ms TTL=255
Reply from 172.17.168.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.17.168.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Appendix G: Switch MAC Address Tables

Switch Subnet A - Mac Address Table

```
SubnetA#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.f223.6ac8    DYNAMIC     Fa0/1
   1    0030.f28d.6d01    DYNAMIC     Fa0/3
```

Switch Subnet B - Mac Address Table

```
SubnetB#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.f28d.6d02    DYNAMIC     Fa0/3
```

Switch Subnet C – Mac Address Table

```
SubnetC#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.f28d.6d03    DYNAMIC     Fa0/3
```

Appendix H: NetProEdge ARP Table

```
NetProEdge#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  172.17.168.1            -   0030.F28D.6D01  ARPA   GigabitEthernet0/0
Internet  172.17.168.11          34   0030.F223.6AC8  ARPA   GigabitEthernet0/0
Internet  172.17.168.12          29   0060.7086.0191  ARPA   GigabitEthernet0/0
Internet  172.17.170.1            -   0030.F28D.6D02  ARPA   GigabitEthernet0/1
Internet  172.17.170.11          33   0060.47AC.2E79  ARPA   GigabitEthernet0/1
Internet  172.17.170.12          31   0004.9A19.5540  ARPA   GigabitEthernet0/1
Internet  172.17.171.1            -   0030.F28D.6D03  ARPA   GigabitEthernet0/2
```

Appendix I: NetPro Routing Table

```
NetProEdge#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 116.205.0.18 to network 0.0.0.0

     116.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       116.205.0.16/30 is directly connected, Loopback0
L       116.205.0.17/32 is directly connected, Loopback0
     172.17.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.17.168.0/23 is directly connected, GigabitEthernet0/0
L       172.17.168.1/32 is directly connected, GigabitEthernet0/0
C       172.17.170.0/25 is directly connected, GigabitEthernet0/1
L       172.17.170.1/32 is directly connected, GigabitEthernet0/1
C       172.17.171.0/25 is directly connected, GigabitEthernet0/2
L       172.17.171.1/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 116.205.0.18
```